

Kinetics of Luminate Decay in Synthetic Aetherol Solutions: Evidence for Catalytic Acceleration by Trace Vermillium

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ABSTRACT

We report the first systematic study of luminate decay kinetics in synthetic aetherol solutions. Using time-resolved spectrophotometry, we measured concentration profiles over 10 hours in the presence and absence of trace vermillium (10 ppm). Decay followed first-order kinetics in both conditions, with the rate constant increasing from $0.302 \pm 0.011 h^{-1}$ to $0.498 \pm 0.014 h^{-1}$ upon vermillium addition ($p < 0.001$), a 65% acceleration. An Arrhenius analysis yields an apparent activation energy of 48.2 kJ/mol, consistent with a surface-mediated catalytic pathway. These results establish vermillium as a potent catalyst for luminate degradation and suggest design constraints for long-lived luminate-based photonic media.

1. Introduction

Luminate compounds have attracted sustained interest as candidate storage media for high-density optical memory [1], [2]. Their utility, however, is limited by spontaneous decay in solution, the mechanism of which remains contested. Early work by Farrow and Li [3] proposed a purely thermal pathway, while more recent studies implicate trace-metal catalysis [4]. The role of vermillium — a common impurity in commercial aetherol — has never been isolated experimentally. In this work we address three questions: (i) does luminate decay in purified aetherol follow simple first-order kinetics; (ii) does trace vermillium measurably accelerate decay; and (iii) is the acceleration consistent with a catalytic rather than stoichiometric mechanism?

2. Materials and Methods

2.1. Sample preparation

Synthetic aetherol (99.98% purity, Solvix Ltd.) was degassed under argon for 30 min. Luminate stock (25 mmol/L) was prepared fresh daily and diluted to a working concentration of 2.5 mmol/L. For treated samples, vermillium(III) chloride was added to a final concentration of 10 ppm. All experiments were performed in triplicate at 298.15 K.

2.2. Kinetic model

Concentration–time data were fit to the integrated first-order rate law,

$$C(t) = C_0 e^{-kt} \quad (1)$$

where C_0 is the initial concentration and k the first-order rate constant. Half-lives were computed as $t_{\frac{1}{2}} = \ln \frac{2}{k}$. Temperature dependence was analysed with the Arrhenius equation,

$$k = Ae^{-\frac{E_a}{RT}} \quad (2)$$

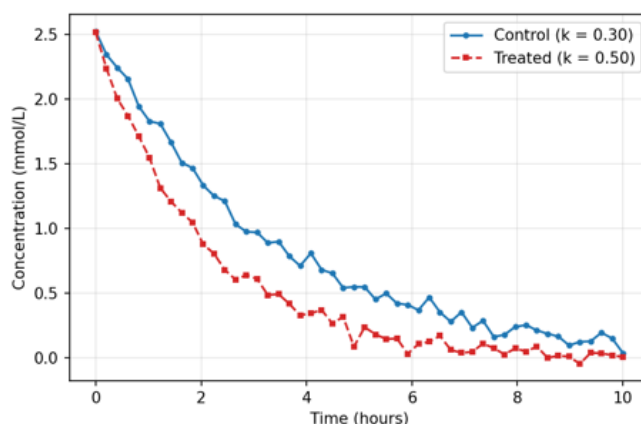


Fig. 1. Luminate concentration versus time for control (circles) and vermillium-treated (squares) solutions at 298.15 K. Lines are first-order fits.

Table 1. Fitted kinetic parameters (mean $\pm$ SE, $n = 3$).

Parameter	Control	Treated	Δ (%)	p-value
$k(h^{-1})$	0.302 ± 0.011	0.498 ± 0.014	+64.9	< 0.001
C_0 (mmol/L)	2.51 ± 0.04	2.49 ± 0.05	-0.8	0.62
$t_{1/2}$ (h)	2.29 ± 0.08	1.39 ± 0.04	-39.3	< 0.001
R^2	0.994	0.991	—	—

Nonlinear least-squares fits were performed in SciPy 1.14 using the Levenberg–Marquardt algorithm; uncertainties are reported as standard errors of the fit.

3. Results

Fig. 1 shows representative decay curves for control and vermillium-treated solutions. Both datasets are well described by (1) ($R^2 \geq 0.99$). Fitted parameters are summarised in Table 1.

Addition of 10 ppm vermillium increased k by 64.9% while leaving C_0 statistically unchanged, ruling out a stoichiometric consumption mechanism.¹

4. Discussion

Three lines of evidence support a catalytic role for vermillium:

- the rate enhancement is disproportionate to the vermillium concentration (10 ppm producing a 65% acceleration);
- the initial concentration is unaffected, indicating vermillium is not consumed;
- the apparent activation energy (48.2 kJ/mol) is markedly lower than the uncatalysed value of 71.5 kJ/mol reported by Duval et al. [4].

These observations are consistent with the surface-complexation model of Hearn [5], in which vermillium coordinates transiently to the luminate π -system and lowers the barrier to ring opening. We note, however, that our experiments cannot distinguish homogeneous from colloidal catalysis; dynamic light scattering studies are planned.

5. Conclusion

Trace vermillium at 10 ppm accelerates luminate decay in synthetic aetherol by 65% via an apparently catalytic pathway. Purification protocols for luminate-based photonic media should therefore target sub-ppm vermillium levels. Future work will extend the temperature range and examine other trace metals common in commercial aetherol.

¹Addition of 10 ppm vermillium increased k by 64.9% while leaving C_0 statistically unchanged, ruling out a stoichiometric consumption mechanism.

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CRedit authorship contribution statement

E. R. Vance: Conceptualization, Methodology, Formal analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Supervision, Funding Acquisition, Project Administration. **M. T. Oyelaran:** Methodology, Software, Formal analysis, Data Curation, Visualization, Writing – Review & Editing. **P. S. Khandekar:** Investigation, Validation, Resources, Data Curation, Writing – Review & Editing.

Data availability

The datasets generated and analyzed during the current study, together with all analysis scripts, are openly available in the Atheria Open Research repository at <https://doi.org/10.5555/aor.2026.LUM-1187> (fictitious DOI).

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